S1.4 Counting particles by mass: The mole

Molar mass (M)

Learning outcomes

By the end of this section you should be able to apply the concepts of the mole and Avogadro's constant, and the definition of molar mass (M), to solve problems involving the relationships between number of particles, amount of substance in moles and mass in grams.

If you have ever followed a recipe, it typically begins with a list of ingredients and the quantities you need for each. The quantities are usually measured by mass or volume and precise measuring will ensure a successful and tasty outcome. Think about the type of equipment you have in your school laboratory – what might you use to measure a mole of a substance? There are no special devices to count one mole of a substance, but there is equipment for measuring the mass and volume, just like in your kitchen (**Interactive 1**).

Rights of use

Interactive 1. Measurements.

More information for interactive 1

An interactive photograph displays various measuring scales and tools used in the kitchen. A weighing scale is placed on a wooden board, with a piece of raw fish on it, displaying a weight of 305 units. Next to the weighing scale, there is a small cup of mayonnaise, a ladle spoon holding a scoop of ice cream, various measuring tablespoons (one containing ghee), a large cup of rice, a 1 by 3 cup of walnuts, a small cup of white kidney beans, a glass container filled milk, and a small glass cup containing sauce.
Beside the wooden board, there is a cup of peas, a conical measuring cup containing flour, an opened bottle with its cap nearby, and a glass of water.
Plus signs, representing interactive hotspots, are present on the weighing scale and the glass container filled with milk. Selecting a hotspot reveals a description for the

corresponding measuring tool.
The hotspot displays the following:
Weighing scale- “Ingredients can be measured by mass in grams or kilograms.”
Glass container- “Ingredients can be measured in volume by millilitres or litres.”
This interactivity helps users recognize the appropriate units and containers used for measuring different types of food items, such as liquids, solids, and semi-solids in everyday cooking.

We cannot count individual elementary entities because they are too small to see and their masses and volumes are too small for any equipment to measure, but we can measure them when they are in a large grouping of 6.02×10^{23} molecules, or one mole. Grouped together makes it possible to measure the mass and volume of a substance, just like when you are following a recipe.

In the previous sections, you learned how to group elementary entities into moles and to assign relative atomic masses to individual atoms, or relative molecular masses for compounds, based on the reference to the carbon-12 isotope. When you are in the school laboratory, chemical substances are easily and accurately measured by the mass, using an analogue or electronic balance. How can we relate the number of elementary entities, grouped and counted as moles, with the carbon-12 isotope reference and measurement on a balance?

We now have to draw a relationship between the amount, in moles, of a substance with its mass. Since each element has a different number of subatomic particles affecting its mass, the relative atomic masses are used to determine the mass of one mole of a substance, known as the molar mass, abbreviated as M. Calculations with molar mass allow us to convert the number of particles or moles of a substance into its mass in grams. The magnitude of molar mass is the same as the relative atomic mass but has the units of grams per mole (g mol^{-1}). For instance, carbon, C, has a relative atomic mass of 12.01 and its molar mass is 12.01 g mol^{-1}.

Concept

Many times, we can infer the units of a new magnitude from its own definition. For example, if we define speed as distance travelled per unit of time, then we can easily argue the possible units to be metres per second, kilometres per hour, etc.

In this section, we have defined molar mass (M) as the mass of one mole of a substance in grams. It is obvious, then, that the units of M are grams per mole (g mol^{-1}).

Can you infer other units, like the units of density (defined as amount of matter per unit of volume)?

Note: when determining units, consider that there is more than one correct combination (metres per second, kilometres per hour, miles per day...) but only one of it is accepted by the international community — the SI unit for speed is metres per second ($m\,s^{-1}$). In your calculations, you will frequently be asked to use SI units, especially in the answers. Unit conversion is an important skill for solving problems in chemistry.

Try some worked examples to check your understanding.

Worked example 1

What is the molar mass, *M*, of the element beryllium, Be?

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involve an element.
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{Be})$: 9.01
Step 3: state the answer with appropriate units and the number of significant figures used in rounding	The molar mass M(I $9.01\ \text{g mol}^{-1}$, becau A_r, is 9.01.

Worked example 2

What is the molar mass, *M*, of nitrogen, N_2?

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding the r
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\mathrm{N})$: 14.01
Step 3: Identify the equation used to find the relative atomic mass.	relative molecular mass $M_r(\mathrm{N_2})$ =
Step 4: substitute the values into the formula.	$M_r(\mathrm{N_2}) = 2 \times 14.01$
Step 5: Find the relative atomic mass.	$= 28.02$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	The molar mass $M(\mathrm{N_2})$ of nitroge because its relative molecular mas

Worked example 3

What is the molar mass, M, of water, H_2O?

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves findin element.
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\mathrm{H})$: 1.01 $A_r(\mathrm{O})$: 16.00
Step 3: Identify the equation used to find the relative atomic mass.	$M_r(\mathrm{H_2O}) = 2\times \mathrm{A_r(H)} + \mathrm{A_r(}$
Step 4: substitute the values into the formula.	$M_r(\mathrm{H_2O}) = (2\times 1.01) + 16.$
Step 5: Find the relative atomic mass.	$= 18.02$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	The molar mass $M(\mathrm{H_2O})$ of because its relative molecul

The molar mass is used when we need to determine the amount (in mol) of a substance from its mass in grams or vice versa. We can use the equation shown below in **Interactive 2**.

$$n = \frac{m}{M}$$

Interactive 2. Mole Equation to Calculate the Amount from a Given Mass of a Substance.

More information for interactive 2

The interactive represents a mole equation which is presented as follows:

$n = \frac{m}{M}$

The letters n, m, and M in the equation, each have a plus icon representing a hotspot. Clicking on the hotspots reveals the corresponding term for each symbol. The hotspots and the corresponding terms are as follows:
Hotspot for n: moles
Hotspot for m: mass
Hotspot for M: molar mass
This equation helps users understand the relationship between the moles, mass, and molar mass, along with their corresponding symbols.

Worked example 4

Determine the mass in grams of 3.00 moles of beryllium, Be.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding element.
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{Be})$: 9.01
Step 3: Find the molar mass.	The relative atomic mass, A_r, molar mass is $9.01\ \text{g mol}^{-1}$. T mass of one mole of beryllium
Step 4: Identify the equation used to find the relative atomic mass.	$n = \frac{m}{M}$
Step 5: Rearrange the formula to make *m* the subject.	$m = n \times M$
Step 6: substitute the values into the formula.	$m = 3.00 \times 9.01$
Step 7: state the answer with appropriate units and the number of significant figures used in rounding.	$= 27.0\ \text{g}$

Worked example 5

Calculate the amount (in mol) in 15.00 grams of sodium chloride, NaCl.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding the mo
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\mathrm{Na})$: 22.99 $A_r(\mathrm{Cl})$: 35.45
Step 3: Find the molar mass.	$M_r(\mathrm{NaCl}) = A_r(\mathrm{Na}) + A_r(\mathrm{Cl})$
Step 4: substitute the values into the formula.	$M_r = 22.99 + 35.45$ $= 58.44$
Step 5: Find the molar mass.	The relative molecular mass, M_r, of N molar mass is $58.44\ \mathrm{g\ mol^{-1}}$. That va one mole of sodium chloride.
Step 6: Identify the equation used to find the number of moles.	$n = \frac{m}{M}$
Step 7: substitute the values into the formula.	$n = \frac{15.00}{58.44}$
Step 8: state the answer with appropriate units and the number of significant figures used in rounding	$= 0.2567$ moles of NaCl

You might be wondering how the quantity of a substance measured in mass relates to the number of elementary entities. You cannot directly determine the number of elementary entities from its mass in grams. You must first convert the mass to an amount in moles, then use Avogadro's constant to determine the number of atoms, molecules, ions or electrons (**Figure 1**).

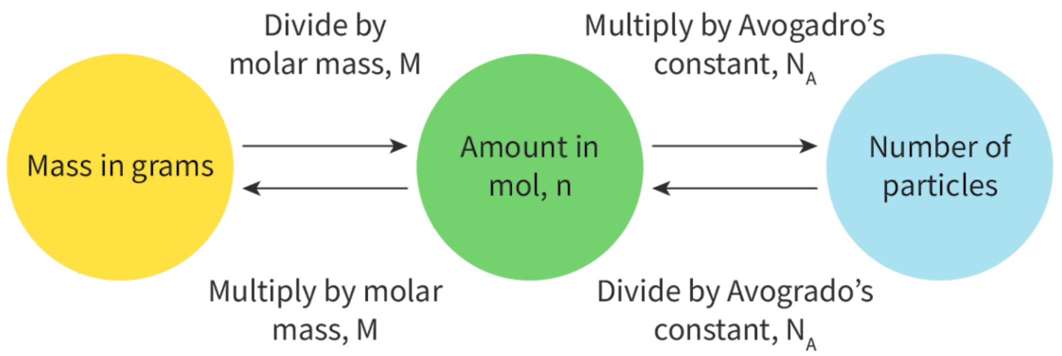

Figure 1. Relationship between the number of moles, the mass in grams and the number of particles of a substance.

More information for figure 1

The diagram illustrates the relationship between mass in grams, the amount in moles, and the number of particles. It uses three circles labeled: "Mass in grams," "Amount in mol, n," and "Number of particles." Arrows indicate the conversions between these quantities, with instructions provided:

1. From "Mass in grams" to "Amount in mol, n": Divide by molar mass, M.
2. From "Amount in mol, n" to "Number of particles": Multiply by Avogadro's constant, N(_A).
3. From "Number of particles" back to "Amount in mol, n": Divide by Avogadro's constant, N(_A).
4. From "Amount in mol, n" back to "Mass in grams": Multiply by molar mass, M.

The circles and text boxes visually demonstrate the flow and connections necessary for calculating between these fundamental chemistry concepts.

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Worked example 6

Given that the molar mass, M, of CO_2 is 44.01 g mol^{-1}, calculate the number of molecules in 75.0 grams of carbon dioxide, CO_2.

Solution steps	Calculations
Step 1: Identify the equation used to find the number of moles.	$n = \frac{m}{M}$
Step 2: substitute the values into the formula.	$n = \frac{75}{44.01}$
Step 3: Find the amount (in mol).	$n = 1.70 \text{ moles of } CO_2$
Step 4: Identify the equation used to convert the number of moles into the number of molecules.	$\text{Number of molecules} = \text{number of}$ ı
Step 5: substitute the values into the formula.	$\text{Number of molecules} = 1.70 \times 6.02 \times$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	$= 1.02 \times 10^{24} \text{ molecules of } CO_2$

Worked example 7

Given that the molar mass, *M*, of Fe is 55.85 g mol^{-1}, calculate the mass of 1.07×10^{25} atoms of iron, Fe.

Solution steps	Calculations
Step 1: Identify the equation used to find the number of moles.	$\text{Number of atoms} = \text{number of}$ ı

Solution steps	Calculations
Step 2: Rearrange the formula to make number of moles the subject.	$\text{amount (in mol)} = \frac{\text{number of a}}{\text{Avogadro co}}$
Step 3: substitute the values into the formula.	$\text{number of moles} = \frac{1.07 \times 10^{25}}{6.02 \times 10^{23}}$ $= 17.77 \text{ moles}$
Step 4: Identify the equation used to convert the number of moles into the mass of the atoms.	$n = \frac{m}{M}$
Step 5: Rearrange to make *m* the subject.	$m = M \times n$
Step 6: substitute the values into the formula.	$m = 55.85 \times 17.77$
Step 7: state the answer with appropriate units and the number of significant figures used in rounding.	$= 992.5 \text{ g of Fe}$

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Research skills — Using search engines and libraries effectively
- **Time required to complete activity:** 25—30 minutes
- **Activity type:** Pair activity

Throughout history and in different industries, there have been many different units of measurement. Here are some examples:

1. A unit of measurement used in the jewellery industry for gemstones is the carat (ct). A 1 carat diamond is equal to a mass of 0.200 grams. Assuming a diamond is made up only of carbon atoms, determine the number of carbon atoms in a 1 carat diamond.
2. A coal sack (sck) was a standardised unit of measurement equivalent to 112 avoirdupois pounds, which is 51 kilograms. Assuming that coal is made up only of carbon atoms, determine the amount in moles and the number of carbon atoms in 1 sack of coal.

Research another unit of measurement used for the mass of a pure substance and calculate the amount in mol and the number of elementary entities. Make a one page infographic to present your findings to your class. Learn about what an infographic (https://youtu.be/zvmDi82xEMc) is and design one using a program such as Canva (https://www.canva.com/).